which represents two particles, or two quanta of the field, interacting. The particle on the left splits into two particles, one of which splits in turn into two particles, which then reunite, converging with the particle on the right. The graph portrays a history of the field's quanta.

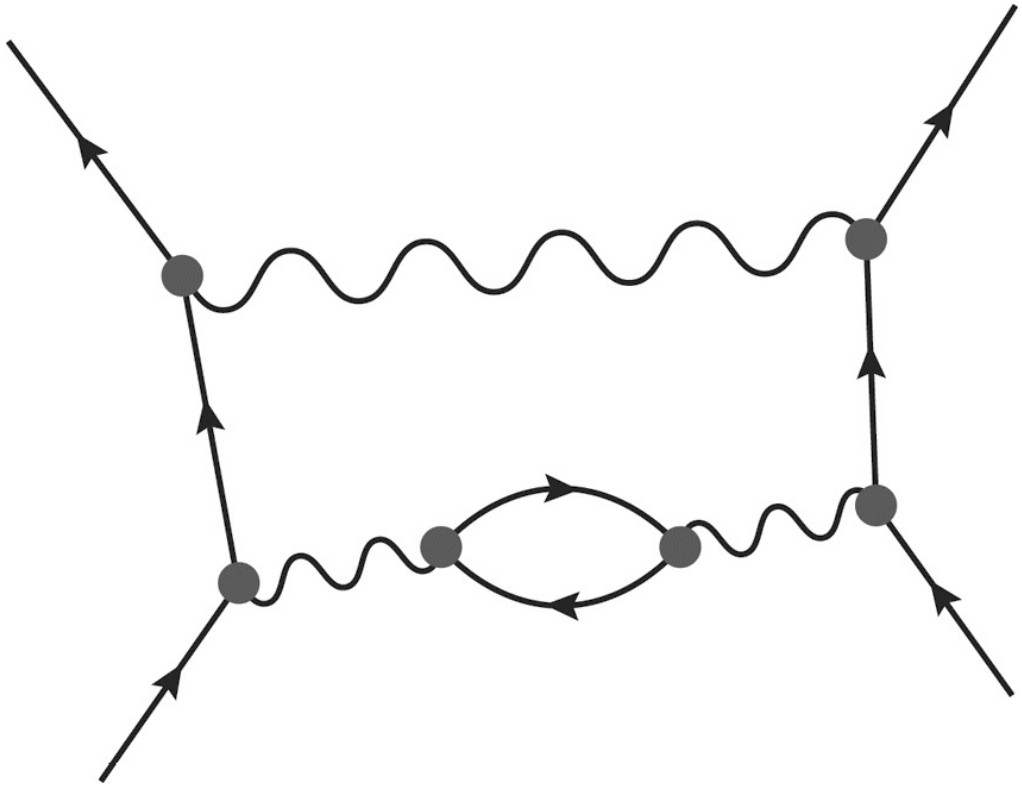


Figure 7.5 A Feynman diagram.

The lattice approximation is used when the forces are strong and the particle picture is no longer effective for describing physics, for instance in computing the strong forces between quarks inside the nucleus of an atom. The lattice technique entails approximating a continuous physical space by means of a lattice, or a grid, as in figure 7.6. This grid is not assumed to be a faithful description of space, but only an approximation, as when engineers calculate the resistance of a bridge by approximating the concrete with a finite number of elements. These two methods of making calculations – Feynman diagrams and the lattice – are the two most efficient techniques of quantum field theory.